The Wave Kinetic Equation

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22nd October 2009

We now derive the kinetic equation for the wave in presence of scattering by density and magnetic field fluctuations. The incident wave is treated as a monochromatic wave, satisfying the wave equation:

$$\nabla \times \nabla \times \mathbf{E} + \varepsilon_0 \mu_0 \frac{\partial^2 \mathbf{E}}{\partial \mathrm{t}^2} = -\mu_0 \frac{\partial \mathbf{J}}{\partial \mathrm{t}} \tag{1}$$

The current $\mathbf{J}$, in absence of extern sources, considering the plasma as a linear medium, is due to the linear response of the plasma to the incident wave field, in the Fourier space the wave equation becomes:

$$\mathbf{k} \times \mathbf{k} \times \mathbf{E}_{\mathbf{k}}(\mathbf{k}, \omega) + \varepsilon_0 \mu_0 \omega^2 \mathbf{E}_{\mathbf{k}}(\mathbf{k}, \omega) = -i \mu_0 \omega \mathbf{J}_{\mathbf{k}}(\mathbf{k}, \omega) \tag{2}$$

In the linear theory we have $\mathbf{J}_{\mathbf{k}}(\mathbf{k}, \omega) = \mathbb{S}(\mathbf{k}, \omega) \cdot \mathbf{E}_{\mathbf{k}}(\mathbf{k}, \omega)$, where $\mathbb{S}(\mathbf{k}, \omega)$is the conductivity tensor, so:

$$\mathbf{k} \times \mathbf{k} \times \mathbf{E}_{\mathbf{k}}(\mathbf{k}, \omega) + \varepsilon_0 \mu_0 \omega^2 \mathbf{E}_{\mathbf{k}}(\mathbf{k}, \omega) = -i \mu_0 \omega \mathbb{S}(\mathbf{k}, \omega) \cdot \mathbf{E}_{\mathbf{k}}(\mathbf{k}, \omega) \tag{3}$$

Introducing the dielectric tensor $\mathbb{K}(\mathbf{k}, \omega) = \mathbb{I} + \frac{i}{\varepsilon_0 \omega} \mathbb{S}(\mathbf{k}, \omega)$ and remembering that $\mathrm{c} = \frac{1}{\sqrt{\varepsilon_0 \mu_0}}$, we obtain:

$$\mathbf{k} \times \mathbf{k} \times \mathbf{E}_{\mathbf{k}}(\mathbf{k}, \omega) + \frac{\omega^2}{\mathrm{c}^2} \mathbb{K}(\mathbf{k}, \omega) \cdot \mathbf{E}_{\mathbf{k}}(\mathbf{k}, \omega) = 0 \tag{4}$$

This is the homogeneous wave equation, that can be expressed in the form:

$$\mathbb{D}(\mathbf{k}, \omega) \cdot \mathbf{E}_{\mathbf{k}}(\mathbf{k}, \omega) = 0 \tag{5}$$

Where $\mathbb{D}(\mathbf{k}, \omega)$ is the dispersion tensor:

$$\mathbb{D}(\mathbf{k}, \omega) = \frac{\mathrm{c}^2}{\omega^2} \left(\mathbf{k}\mathbf{k} - k^2 \mathbb{I} \right) + \mathbb{K}(\mathbf{k}, \omega) \tag{6}$$

If we have fluctuations of density and magnetic field, $\delta n\left(\widetilde{\mathbf{k}}, \widetilde{\omega}\right)$ and $\delta \mathbf{B}\left(\widetilde{\mathbf{k}}, \widetilde{\omega}\right)$, from the conservation of momentum and energy, the wave vector and the frequency of the scattered wave are given by the expressions:

$$\check{\mathbf{k}} = \mathbf{k} + \widetilde{\mathbf{k}}, \quad \check{\omega} = \omega + \widetilde{\omega} \tag{7}$$

The field of the scattered wave, $\check{\mathbf{E}}\left(\check{\mathbf{k}}, \check{\omega}\right)$, satisfies the equation:

$$\nabla \times \nabla \times \check{\mathbf{E}} + \varepsilon_0 \mu_0 \frac{\partial^2 \check{\mathbf{E}}}{\partial \mathrm{t}^2} = -\mu_0 \frac{\partial}{\partial \mathrm{t}} \left(\check{\mathbf{J}} + \widetilde{\mathbf{J}}\right) \tag{8}$$

Where $\check{\mathbf{J}}$ is the current due to the linear response of the plasma to the field of the scattered wave, and $\widetilde{\mathbf{J}}$ is the current due to the interaction of the incident wave with the fluctuations. In the Fourier space we have:

$$\check{\mathbf{k}} \times \check{\mathbf{k}} \times \check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}}, \check{\omega}\right) + \varepsilon_0 \mu_0 \check{\omega}^2 \check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}}, \check{\omega}\right) = -i\mu_0 \check{\omega} \left(\check{\mathbf{J}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}}, \check{\omega}\right) + \widetilde{\mathbf{J}}_{\mathbf{k},\check{\mathbf{k}}}\left(\mathbf{k}.\check{\mathbf{k}}, \omega, \check{\omega}\right)\right) \tag{9}$$

So, $\check{\mathbf{J}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}}, \check{\omega}\right)$ is given by:

$$\check{\mathbf{J}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}}, \check{\omega}\right) = \mathbb{S}\left(\check{\mathbf{k}}, \check{\omega}\right) \cdot \check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}}, \check{\omega}\right) \tag{10}$$

Inserting this expression in the wave equation and introducing the dielectric tensor, we obtain the non homogeneous equation:

$$\check{\mathbf{k}} \times \check{\mathbf{k}} \times \check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}}, \check{\omega}\right) + \frac{\check{\omega}^2}{\mathrm{c}^2} \mathbb{K}\left(\check{\mathbf{k}}, \check{\omega}\right) \cdot \check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}}, \check{\omega}\right) = -i\mu_0 \check{\omega} \widetilde{\mathbf{J}}_{\mathbf{k},\check{\mathbf{k}}}\left(\mathbf{k}, \check{\mathbf{k}}, \omega, \check{\omega}\right) \tag{11}$$

Using the dispersion tensor the equation becomes:

$$\mathbb{D}\left(\check{\mathbf{k}}, \check{\omega}\right) \cdot \check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}}, \check{\omega}\right) = -\frac{i}{\varepsilon_0 \check{\omega}} \widetilde{\mathbf{J}}_{\mathbf{k},\check{\mathbf{k}}}\left(\mathbf{k}, \check{\mathbf{k}}, \omega, \check{\omega}\right) \tag{12}$$

An explicit expression for the current $\widetilde{\mathbf{J}}_{\mathbf{k},\check{\mathbf{k}}}\left(\mathbf{k}, \check{\mathbf{k}}, \omega, \check{\omega}\right)$ is obtained applying the Fourier trasform to the MHD equations, the derivation of this expression using the linear theory in the cold plasma approximation is discussed in details in appendix. In the cold plasma limit, considering only the component of the current due to the electrons, and several incident modes with wave vector $\mathbf{k}$ and frequency ω, we have:

$$\widetilde{\mathbf{J}}_{\mathbf{k},\check{\mathbf{k}}}\left(\mathbf{k}, \check{\mathbf{k}}, \omega, \check{\omega}\right) = i\varepsilon_0 \sum_{\mathbf{k},\omega} \omega \left[\left(\mathbb{I} - \mathbb{K}\left(\mathbf{k}, \omega\right)\right) \frac{\delta n\left(\check{\mathbf{k}} - \mathbf{k}, \check{\omega} - \omega\right)}{n} + \right.$$

$$\left. + i \frac{\omega_{ce} \check{\omega}}{\omega_{pe}^2} \left(\mathbb{I} - \mathbb{K}\left(\check{\mathbf{k}}, \check{\omega}\right)\right) \cdot \left(\mathbb{E}^{LC} \cdot \frac{\delta \mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}} - \mathbf{k}, \check{\omega} - \omega\right)}{B_{\mathbf{k}}\left(\mathbf{k}, \omega\right)} \right) \cdot \left(\mathbb{I} - \mathbb{K}\left(\mathbf{k}, \omega\right)\right) \right] \cdot \mathbf{E}_{\mathbf{k}}\left(\mathbf{k}, \omega\right) \tag{13}$$

Where $\mathbb{E}^{LC}$ is the Levi-Civita tensor of order 3, and $\mathbb{K}$ is the cold plasma dielectric tensor. Using this expression for the current due to the fluctuations, the wave equation becomes:

$$\mathbb{D}\left(\check{\mathbf{k}},\breve{\omega}\right)\cdot\check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}}\breve{\omega}\right)=\sum_{\mathbf{k},\omega}\left[\frac{\omega}{\breve{\omega}}\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k},\omega\right)\right)\frac{\delta n\left(\check{\mathbf{k}}-\mathbf{k},\breve{\omega}-\omega\right)}{n}+\right.$$

$$\left.+i\frac{\omega_{ce}\omega}{\omega_{pe}^{2}}\left(\mathbb{I}-\mathbb{K}\left(\check{\mathbf{k}},\breve{\omega}\right)\right)\cdot\left(\mathbb{E}^{LC}\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k},\breve{\omega}-\omega\right)}{B_{\mathbf{k}}\left(\mathbf{k},\omega\right)}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k},\omega\right)\right)\right]\cdot\mathbf{E}_{\mathbf{k}}\left(\mathbf{k},\omega\right) \tag{14}$$

Fluctuations are assumed to be slow, so we have $\widetilde{\omega}\ll\omega$ and $\breve{\omega}\approx\omega$, as a consequence:

$$\mathbb{D}\left(\check{\mathbf{k}},\breve{\omega}\right)\cdot\check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}},\breve{\omega}\right)=\sum_{\mathbf{k},\omega}\left[\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k},\omega\right)\right)\frac{\delta n\left(\check{\mathbf{k}}-\mathbf{k}\right)}{n}+\right.$$

$$\left.+i\frac{\omega_{ce}\omega}{\omega_{pe}^{2}}\left(\mathbb{I}-\mathbb{K}\left(\check{\mathbf{k}},\breve{\omega}\right)\right)\cdot\left(\mathbb{E}^{LC}\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k}\right)}{B_{\mathbf{k}}\left(\mathbf{k},\omega\right)}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k},\omega\right)\right)\right]\cdot\mathbf{E}_{\mathbf{k}}\left(\mathbf{k},\omega\right) \tag{15}$$

Now, we assume that fluctuations of the density and of the magnetic field are sufficienly small to be treated with a perturbative approach, it means that the effects of the fluctuations may be rapresented by a small variation from the normal mode propagation that we can easily study by solving the homogeneous wave equation. If we use this perturbative approach, the frequency of the scattered wave is given by the expression:

$$\breve{\omega}=\breve{\omega}_{\check{k}}+i\frac{\partial}{\partial\mathrm{t}}\qquad\text{with}\ \left|\frac{\partial}{\partial\mathrm{t}}\right|\ll\breve{\omega}_{\check{k}} \tag{16}$$

Where $\breve{\omega}_{\check{k}}$ is the frequency of the normal mode, that is decribed by the homogeneous equation:

$$\mathbb{D}\left(\check{\mathbf{k}},\breve{\omega}_{\check{k}}\right)\cdot\check{\mathbf{E}}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}},\breve{\omega}_{\check{k}}\right)=0 \tag{17}$$

The fluctuations add a term to the imaginary part of the frequency, this means that there is a term due to the fluctuations in the energy balance equation (δn and $\delta\mathbf{B}$ give an additional contribution to the damping of the wave). $\breve{\omega}_{\check{k}}$ can be easily determined by solving the dispersion relation associated with the homogeneous equation. Inserting the expression for $\breve{\omega}$ in the wave equation, and expressing the electric field in function of the polarization vector $\mathbf{E}(\mathbf{k})=\mathbf{e}\left(\mathbf{k}\right)E\left(\mathbf{k}\right)$, we obtain:

$$\left[\mathbb{D}\left(\check{\mathbf{k}},\breve{\omega}_{\check{k}}\right)+i\frac{\partial\mathbb{D}}{\partial\breve{\omega}}\frac{\partial}{\partial\mathrm{t}}\right]\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\check{E}_{\check{\mathbf{k}}}\left(\check{\mathbf{k}}\right)=\sum_{\mathbf{k}}\left[\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k}\right)\right)\frac{\delta n\left(\check{\mathbf{k}}-\mathbf{k}\right)}{n}+\right.$$

$$\left.+i\frac{\omega_{ce}\omega}{\omega_{pe}^{2}}\left(\mathbb{I}-\mathbb{K}\left(\check{\mathbf{k}}\right)\right)\cdot\left(\mathbb{E}^{LC}\cdot\frac{\delta\mathbf{B}_{\check{\mathbf{k}}-\mathbf{k}}\left(\check{\mathbf{k}}-\mathbf{k}\right)}{B_{\mathbf{k}}\left(\mathbf{k}\right)}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k}\right)\right)\right]\cdot\mathbf{e}\left(\mathbf{k}\right)E_{\mathbf{k}}\left(\mathbf{k}\right) \tag{18}$$

In this equation all the quantities are considered for $\omega = \breve{\omega}_{\breve{k}}$. Multiplying this equation for $\breve{\mathbf{e}}^{\star}\left(\breve{\mathbf{k}}\right)$ and remembering that $\breve{\mathbf{e}}^{\star}\left(\breve{\mathbf{k}}\right) \cdot \mathbb{D}\left(\breve{\mathbf{k}}, \breve{\omega}_{\breve{k}}\right) = 0$, the equation finally becomes:

$$i\breve{\mathbf{e}}^{\star}\left(\breve{\mathbf{k}}\right) \cdot \frac{\partial \mathbb{D}}{\partial \breve{\omega}} \cdot \breve{\mathbf{e}}\left(\breve{\mathbf{k}}\right) \frac{\partial \breve{E}_{\breve{\mathbf{k}}}\left(\breve{\mathbf{k}}\right)}{\partial \mathrm{t}} = \sum_{\mathbf{k}} \breve{\mathbf{e}}^{\star}\left(\breve{\mathbf{k}}\right) \cdot \left[\left(\mathbb{I} - \mathbb{K}\left(\mathbf{k}\right)\right) \frac{\delta n\left(\breve{\mathbf{k}} - \mathbf{k}\right)}{n} + \right.$$

$$\left. + i\frac{\omega_{ce}\omega}{\omega_{pe}^2}\left(\mathbb{I} - \mathbb{K}\left(\breve{\mathbf{k}}\right)\right) \cdot \left(\mathbb{E}^{LC} \cdot \frac{\delta \mathbf{B}_{\breve{\mathbf{k}}-\mathbf{k}}\left(\breve{\mathbf{k}} - \mathbf{k}\right)}{B_{\mathbf{k}}\left(\mathbf{k}\right)} \right) \cdot \left(\mathbb{I} - \mathbb{K}\left(\mathbf{k}\right)\right) \right] \cdot \mathbf{e}\left(\mathbf{k}\right) E_{\mathbf{k}}\left(\mathbf{k}\right) \quad (19)$$

In order to study the coupling between the incident and the scattered wave, and expecially the enegy exchange between the two modes, we introduce the wave energy density:

$$W\left(\mathbf{k}\right) = \varepsilon_0 \frac{\breve{\omega}_{\breve{k}}}{2} \mathbf{e}^{\star}\left(\mathbf{k}\right) \cdot \frac{\partial \mathbb{D}}{\partial \omega} \cdot \mathbf{e}\left(\mathbf{k}\right) \left|\mathbf{E}\left(\mathbf{k}\right)\right|^2 = \varepsilon_0 \mathbf{e}^{\star}\left(\mathbf{k}\right) \cdot \mathbb{M} \cdot \mathbf{e}\left(\mathbf{k}\right) E\left(\mathbf{k}\right)^2 \quad (20)$$

Where $\mathbb{M} = \frac{\breve{\omega}_{\breve{k}}}{2} \frac{\partial \mathbb{D}}{\partial \omega}$. The wave amplitude is given by:

$$C\left(\mathbf{k}\right) = W\left(\mathbf{k}\right)^{\frac{1}{2}} \quad (21)$$

So we exprime the electric field in function of the wave amplitude, and we obtain the mode coupling equation:

$$i\frac{\partial C\left(\breve{\mathbf{k}}\right)}{\partial \mathrm{t}} = \sum_{\mathbf{k}} \frac{\breve{\omega}_{\breve{k}}}{2} \left[\frac{\breve{\mathbf{e}}^{\star}\left(\breve{\mathbf{k}}\right) \cdot \left(\mathbb{I} - \mathbb{K}\left(\mathbf{k}\right)\right) \cdot \mathbf{e}\left(\mathbf{k}\right)}{\left(\breve{\mathbf{e}}^{\star}\left(\breve{\mathbf{k}}\right) \cdot \mathbb{M} \cdot \breve{\mathbf{e}}\left(\breve{\mathbf{k}}\right)\right)^{\frac{1}{2}} \left(\mathbf{e}^{\star}\left(\mathbf{k}\right) \cdot \mathbb{M} \cdot \mathbf{e}\left(\mathbf{k}\right)\right)^{\frac{1}{2}}} \frac{\delta n\left(\breve{\mathbf{k}} - \mathbf{k}\right)}{n} + \right.$$

$$\left. + i\frac{\omega_{ce}\breve{\omega}_{\breve{k}}}{\omega_{pe}^2} \frac{\breve{\mathbf{e}}^{\star}\left(\breve{\mathbf{k}}\right) \cdot \left(\mathbb{I} - \mathbb{K}\left(\breve{\mathbf{k}}\right)\right) \cdot \left(\mathbb{E}^{LC} \cdot \frac{\delta \mathbf{B}_{\breve{\mathbf{k}}-\mathbf{k}}\left(\breve{\mathbf{k}} - \mathbf{k}\right)}{B_{\mathbf{k}}\left(\mathbf{k}\right)} \right) \cdot \left(\mathbb{I} - \mathbb{K}\left(\mathbf{k}\right)\right) \cdot \mathbf{e}\left(\mathbf{k}\right)}{\left(\breve{\mathbf{e}}^{\star}\left(\breve{\mathbf{k}}\right) \cdot \mathbb{M} \cdot \breve{\mathbf{e}}\left(\breve{\mathbf{k}}\right)\right)^{\frac{1}{2}} \left(\mathbf{e}^{\star}\left(\mathbf{k}\right) \cdot \mathbb{M} \cdot \mathbf{e}\left(\mathbf{k}\right)\right)^{\frac{1}{2}}} \right] C\left(\mathbf{k}\right) \quad (22)$$

This equation describes the evolution of the wave amplitude, and so of the wave energy, due to the scattering by fluctuations, it can be written in the form:

$$i\frac{\partial C\left(\breve{\mathbf{k}}\right)}{\partial \mathrm{t}} = \sum_{\mathbf{k}} \left[V^n\left(\mathbf{k}, \breve{\mathbf{k}}, \breve{\omega}_{\breve{k}}\right) \frac{\delta n\left(\breve{\mathbf{k}} - \mathbf{k}\right)}{n} + \mathbf{V}^{\mathbf{B}}\left(\mathbf{k}, \breve{\mathbf{k}}, \breve{\omega}_{\breve{k}}\right) \cdot \frac{\delta \mathbf{B}_{\breve{\mathbf{k}}-\mathbf{k}}\left(\breve{\mathbf{k}} - \mathbf{k}\right)}{B_{\mathbf{k}}} \right] C\left(\mathbf{k}\right) \quad (23)$$

Where $V^n\left(\mathbf{k}, \breve{\mathbf{k}}, \omega_k\right)$and $\mathbf{V}^{\mathbf{B}}\left(\mathbf{k}, \breve{\mathbf{k}}, \omega_k\right)$ are the coupling coefficients:

$$V^n\left(\mathbf{k}, \breve{\mathbf{k}}, \omega_k\right) = \frac{\breve{\omega}_{\breve{k}}}{2} \frac{\breve{\mathbf{e}}^{\star}\left(\breve{\mathbf{k}}\right) \cdot \left(\mathbb{I} - \mathbb{K}\left(\mathbf{k}\right)\right) \cdot \mathbf{e}\left(\mathbf{k}\right)}{\left(\breve{\mathbf{e}}^{\star}\left(\breve{\mathbf{k}}\right) \cdot \mathbb{M} \cdot \breve{\mathbf{e}}\left(\breve{\mathbf{k}}\right)\right)^{\frac{1}{2}} \left(\mathbf{e}^{\star}\left(\mathbf{k}\right) \cdot \mathbb{M} \cdot \mathbf{e}\left(\mathbf{k}\right)\right)^{\frac{1}{2}}} \quad (24)$$

$$\mathbf{V}^{\mathbf{B}}\left(\mathbf{k},\check{\mathbf{k}},\omega_k\right)=\frac{i\omega_{ce}\check{\omega}_{\check{k}}^2}{2\omega_{pe}^2}\frac{\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\left(\mathbb{I}-\mathbb{K}\left(\check{\mathbf{k}}\right)\right)\cdot\mathbb{E}^{LC}\cdot\left(\mathbb{I}-\mathbb{K}\left(\mathbf{k}\right)\right)\cdot\mathbf{e}\left(\mathbf{k}\right)}{\left(\check{\mathbf{e}}^{\star}\left(\check{\mathbf{k}}\right)\cdot\mathbb{M}\cdot\check{\mathbf{e}}\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\left(\mathbf{e}^{\star}\left(\mathbf{k}\right)\cdot\mathbb{M}\cdot\mathbf{e}\left(\mathbf{k}\right)\right)^{\frac{1}{2}}} \tag{25}$$

Using the component notation:

$$V^{n}\left(\mathbf{k},\check{\mathbf{k}},\omega_k\right)=\frac{\check{\omega}_{\check{k}}}{2}\frac{\check{e}_i^{\star}\left(\check{\mathbf{k}}\right)\left(\delta_{ij}-K_{ij}\left(\mathbf{k}\right)\right)e_j\left(\mathbf{k}\right)}{\left(\check{e}_i^{\star}\left(\check{\mathbf{k}}\right)M_{ij}\check{e}_j\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\left(e_k^{\star}\left(\mathbf{k}\right)M_{kl}e_l\left(\mathbf{k}\right)\right)^{\frac{1}{2}}} \tag{26}$$

$$V_m^{\mathbf{B}}\left(\mathbf{k},\check{\mathbf{k}},\omega_k\right)=\frac{i\omega_{ce}\check{\omega}_{\check{k}}^2}{2\omega_{pe}^2}\frac{\check{e}_i^{\star}\left(\check{\mathbf{k}}\right)\left(\delta_{il}-K_{il}\left(\check{\mathbf{k}}\right)\right)\epsilon_{lkm}\left(\delta_{kj}-K_{kj}\left(\mathbf{k}\right)\right)e_j\left(\mathbf{k}\right)}{\left(\check{e}_i^{\star}\left(\check{\mathbf{k}}\right)M_{ij}\check{e}_j\left(\check{\mathbf{k}}\right)\right)^{\frac{1}{2}}\left(e_k^{\star}\left(\mathbf{k}\right)M_{kl}e_l\left(\mathbf{k}\right)\right)^{\frac{1}{2}}} \tag{27}$$

Appendix A

We now derive the explicit expression for the current $\widetilde{\mathbf{J}}_{\mathbf{k},\check{\mathbf{k}}}\left(\mathbf{k},\check{\mathbf{k}},\omega,\check{\omega}\right)$ due to interaction between the field of the incident wave and the fluctuation of the density and of the magnetic field, in the cold plasma approximation.

We firstly assume to have only fluctuations of the density, with $\widetilde{\omega}\ll\omega$, so $\check{\omega}=\omega$. In the cold plasma model, the effects of the thermal motion of the particles and of the collisions are neglected, each species of particle is treated as an indipendent fluid, so for studying wave propagation we need to solve the Maxwell equations coupled with the equation of the motion for the various species of particles:

$$\nabla\times\mathbf{E}=-\frac{\partial\mathbf{B}}{\partial\mathrm{t}} \tag{28}$$

$$\nabla\times\mathbf{B}=\varepsilon_0\mu_0\frac{\partial\mathbf{E}}{\partial\mathrm{t}}+\mu_0\mathbf{J} \tag{29}$$

$$\mathrm{m}_s\left(\frac{\partial\mathbf{v}_s}{\partial\mathrm{t}}+\left(\mathbf{v}_s\nabla\right)\mathbf{v}_s\right)=\mathrm{q}_s\left(\mathbf{E}+\frac{\mathbf{v}_s}{\mathrm{c}}\times\mathbf{B}\right) \tag{30}$$

Where the current $\mathbf{J}$ is given by the expression:

$$\mathbf{J}=\sum_s\mathrm{q}_s n_s\mathbf{v}_s \tag{31}$$

Since the difference between electron and ion masses is very large, and $\mathbf{v}_i\ll\mathbf{v}_e$, we consider only the component of the current due to the electrons, assuming $n_e=n$ and introducing the density fluctuations, we have:

$$\mathbf{J}=-e\left(n+\delta n\right)\mathbf{v}_e \tag{32}$$

Neglecting the convective term, the linearized equation of motion, in absence of magnetic field and velocity fluctuation, is:

$$\mathrm{m}_e \frac{\partial \mathbf{v}_e}{\partial \mathrm{t}} = -e\left(\mathbf{E} + \frac{\mathbf{v}_e}{\mathrm{c}} \times \mathbf{B}\right) \tag{33}$$

In the Fourier space we have:

$$i\mathrm{m}_e\omega\left[\mathbf{v}_{e\mathbf{k}}(\mathbf{k},\omega) + i\frac{\omega_{ce}}{\omega}\mathbf{v}_{e\mathbf{k}}(\mathbf{k},\omega) \times \mathbf{b}_{\mathbf{k}}(\mathbf{k},\omega)\right] = e\mathbf{E}(\mathbf{k},\omega) \tag{34}$$

Using the tensorial notation:

$$i\mathrm{m}_e\omega\left(\mathbb{I} + i\frac{\omega_{ce}}{\omega}\mathbb{E}^{LC} \cdot \mathbf{b}_{\mathbf{k}}(\mathbf{k},\omega)\right) \cdot \mathbf{v}_{e\mathbf{k}}(\mathbf{k},\omega) = e\mathbf{E}(\mathbf{k},\omega) \tag{35}$$

So the electron velocity is given by the expression:

$$\mathbf{v}_{e\mathbf{k}}(\mathbf{k},\omega) = -\frac{ie}{\mathrm{m}_e\omega}\mathbb{T}(\mathbf{k},\omega) \cdot \mathbf{E}_{\mathbf{k}}(\mathbf{k},\omega) \tag{36}$$

Where the tensor $\mathbb{T}$ is:

$$\mathbb{T}(\mathbf{k},\omega) = \frac{1}{\omega^2 - \omega_{ce}^2}\left(\omega^2\mathbb{I} - \omega_{ce}^2\mathbf{b}_{\mathbf{k}}\mathbf{b}_{\mathbf{k}} - i\omega\omega_{ce}\mathbb{E}^{LC} \cdot \mathbf{b}_{\mathbf{k}}\right)$$

Using (30) in (25), the current becomes:

$$\mathbf{J}_{\mathbf{k}} = \frac{ie^2 n}{\mathrm{m}_e\omega}\left(1 + \frac{\delta n\left(\check{\mathbf{k}} - \mathbf{k}\right)}{n}\right)\mathbb{T}(\mathbf{k},\omega)\cdot\mathbf{E}_{\mathbf{k}}(\mathbf{k},\omega) = \frac{i\omega_{pe}^2\varepsilon_0}{\omega}\left(1 + \frac{\delta n\left(\check{\mathbf{k}} - \mathbf{k}\right)}{n}\right)\mathbb{T}(\mathbf{k},\omega)\cdot\mathbf{E}_{\mathbf{k}}(\mathbf{k},\omega) \tag{37}$$

So the component due to the interaction between the field of the incident wave and the fluctuations is:

$$\widetilde{\mathbf{J}}_{\mathbf{k},\check{\mathbf{k}}}\left(\mathbf{k},\check{\mathbf{k}},\omega,\check{\omega}\right) = \frac{i\omega_{pe}^2\varepsilon_0}{\omega}\frac{\delta n\left(\check{\mathbf{k}} - \mathbf{k}\right)}{n}\mathbb{T}(\mathbf{k},\omega) \cdot \mathbf{E}_{\mathbf{k}}(\mathbf{k},\omega) \tag{38}$$

In absence of fluctuations, we have:

$$\mathbf{J}_{\mathbf{k}}(\mathbf{k},\omega) = \mathbb{S}(\mathbf{k},\omega)\cdot\mathbf{E}_{\mathbf{k}}(\mathbf{k},\omega) = i\omega\varepsilon_0\left(\mathbb{I} - \mathbb{K}(\mathbf{k},\omega)\right)\cdot\mathbf{E}_{\mathbf{k}}(\mathbf{k},\omega) = \frac{i\omega_{pe}^2\varepsilon_0}{\omega}\mathbb{T}(\mathbf{k},\omega)\cdot\mathbf{E}_{\mathbf{k}}(\mathbf{k},\omega) \tag{39}$$

Where $\mathbb{S}$ and $\mathbb{K}$ are respectively the conductivity and the dielectric tensor of a cold plasma. From last equation we can deduce an expression of $\mathbb{T}$ in function of the cold plasma dielectric tensor:

$$\mathbb{T}(\mathbf{k},\omega) = \frac{\omega^2}{\omega_{pe}^2}\left(\mathbb{I} - \mathbb{K}(\mathbf{k},\omega)\right)$$

So the current due to the fluctuation of the density can be finally expressed in the form:

$$\widetilde{\mathbf{J}}_{\mathbf{k},\check{\mathbf{k}}}\left(\mathbf{k},\check{\mathbf{k}},\omega,\check{\omega}\right) = i\omega\varepsilon_0\frac{\delta n\left(\check{\mathbf{k}} - \mathbf{k}\right)}{n}\left(\mathbb{I} - \mathbb{K}(\mathbf{k},\omega)\right) \cdot \mathbf{E}_{\mathbf{k}}(\mathbf{k},\omega) \tag{40}$$

In components:

$$\widetilde{J}_{\mathbf{k},\check{\mathbf{k}},i} = i\omega\varepsilon_0 \left(\delta_{ij} - K_{ij}\left(\mathbf{k},\omega\right)\right) \frac{\delta n\left(\check{\mathbf{k}} - \mathbf{k}\right)}{n} E_{\mathbf{k},j} \tag{41}$$